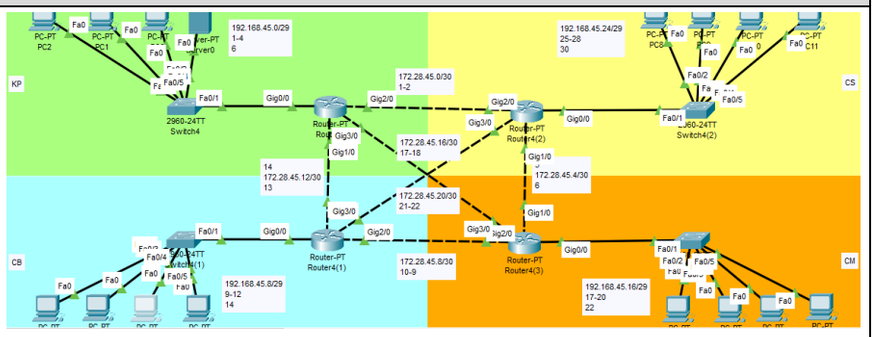
Simulasi Jaringan WAN Perusahaan xyz.net

Perusahaan xyz.net mempunyai kantor dan perangkat sebagai berikut :

Branch	Router	Switch	Max. PC/Server
Kantor Pusat (KP)	1	1	4 (1 server, 3 PC)
Cabang Bandung (CB)	1	1	4 PC
Cabang Makasar (CM)	1	1	4 PC
Cabang Semarang (CS)	1	1	4 PC

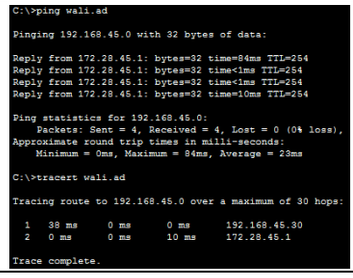
1. Simulasikan dengan software packet tracert
2. Buatlah topologi jaringan dan alokasi IP Address yang loadbalancing, supaya KP bisa terhubung ke CB, CM dan CS. Begitu juga antar cabang juga bisa terhubung.
3. Block IP address antar router 172.28.45.0/24, sedangkan block IP address di PC/Server 192.168.45.0/24 alokasikan ip address tersebut secara efektif efisien.
4. Kantor pusat mempunyai 1 server yang berfungsi DNS server dan webserver, dengan domain wali.id
5. Router CB, CM dan CS mempunyai service DHCP
6. Router bekerja dengan protocol routing dynamic OSPF.
7. Lakukan uji coba **ping** dan **traceroute** dan capture hasilnya:
8. Lakukan Ujicoba ping ke wali.id
9. Tampilkan **configration script** perangkat router KP, CB, CM, dan CS
10. Buat laporan masukan ke Laporan final project praktikum format terlapir.

NO	DESKRIPSI	GAMBAR DAN URAIAN
1	Screen topologi yang dibuat dengan packet tracert sesuai soal diatas dilengkapi dengan dokumentasi IP Adress	 Buatlah topologi seperti yang diperintahkan soal diatas
2	Konfigurasi dan tampilkan IP, subnet, gateway, dan DNS Kantor Pusat (KP)	Cisco Packet Tracer SERVER Command Line 1.0 C:\>ipconfig 192.168.45.1 255.255.255.248 192.168.45.6 C:\>ipconfig FastEthernet0 Connection:(default port) Connection-specific DNS Suffix..: Link-local IPv6 Address.....: FE80::2D0:BAFF:FE34:5971 IPv6 Address.....: :: IPv4 Address.....: 192.168.45.1 Subnet Mask.....: 255.255.255.248 Default Gateway.....: :: 192.168.45.6
3	Konfigurasi dan tampilkan IP, subnet, gateway, dan DNS Cabang Bandung (CB)	C:\>ipconfig /renew IP Address.....: 192.168.45.9 Subnet Mask.....: 255.255.255.248 Default Gateway.....: 192.168.45.14 DNS Server.....: 192.168.45.1
4	Konfigurasi dan tampilkan IP, subnet, gateway, dan DNS Cabang Makasar (CM)	C:\>ipconfig /renew IP Address.....: 192.168.45.17 Subnet Mask.....: 255.255.255.248 Default Gateway.....: 192.168.45.22 DNS Server.....: 192.168.45.1
5	Konfigurasi dan tampilkan IP, subnet, gateway, dan DNS Cabang Semarang (CS)	C:\>ipconfig /renew IP Address.....: 192.168.45.25 Subnet Mask.....: 255.255.255.248 Default Gateway.....: 192.168.45.30 DNS Server.....: 192.168.45.1
6	Konfigurasi dan tampikan IP router Kantor Pusat (KP)	Router>ena Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#int gig 0/0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up Router(config-if)#ip address 192.168.45.0 255.255.255.252 Bad mask /30 for address 192.168.45.0 Router(config-if)#do write Building configuration... [OK] Router(config-if)# Router(config)#int gig 2/0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet2/0, changed state to up Router(config-if)#ip address 172.28.45.1 255.255.252 ^ % Invalid input detected at '^' marker. Router(config-if)#ip address 172.28.45.1 255.255.255.252 Router(config-if)#do write Building configuration... [OK] Router(config-if)# Router(config)#int gig 1/0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet1/0, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0, changed state to up Router(config-if)#ip add 172.28.45.14 255.255.255.252 Router(config-if)#do write Building configuration... [OK] Router(config-if)#

		```Router(config)#int gig 3/0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet3/0, changed state to up Router(config-if)#ip add 172.28.45.22 255.255.255.252 Router(config-if)#do write Building configuration... [OK] Router#show ip int brief Interface IP-Address OK? Method Status Protocol GigabitEthernet0/0 192.168.45.6 YES manual up up GigabitEthernet1/0 172.28.45.14 YES manual up up GigabitEthernet2/0 172.28.45.1 YES manual up up GigabitEthernet3/0 172.28.45.17 YES manual up up Router#```
7	Konfigurasi dan tampikan IP router Cabang Bandung (CB)	```Router>enable Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#int gig 0/0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up Router(config-if)#ip address 192.168.45.14 255.255.255.248 Router(config-if)#do write Building configuration... [OK] Router(config)#int gig 2/0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet2/0, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet2/0, changed state to up Router(config-if)#ip add 172.28.45.10 255.255.255.252 Router(config-if)#do write Building configuration... [OK] Router(config)#int gig 1/0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet1/0, changed state to up Router(config-if)#ip add 172.28.45.13 255.255.255.252 Router(config-if)#do write Building configuration... [OK] Router(config)#int gig 3/0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet3/0, changed state to up Router(config-if)#ip add 172.28.45.22 255.255.255.252 Router(config-if)#do write Building configuration... [OK] Router#show ip int br Interface IP-Address OK? Method Status Protocol GigabitEthernet0/0 192.168.45.14 YES manual up up GigabitEthernet1/0 172.28.45.13 YES manual up up GigabitEthernet2/0 172.28.45.10 YES manual up up GigabitEthernet3/0 172.28.45.22 YES manual up up```
8	Konfigurasi dan tampikan IP router Cabang Makasar (CM)	```Router>enab Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#int gig 0/0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up Router(config-if)#ip address 192.168.45.22 255.255.255.248 Router(config-if)#do write Building configuration... [OK] Router(config)#int gig 1/0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet1/0, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0, changed state to up Router(config-if)#ip address 172.28.45.6 255.255.255.252 Router(config-if)#do write Building configuration... [OK] Router(config)#int gig 2/0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet2/0, changed state to up Router(config-if)#ip address 172.168.45.9 255.255.255.252 Router(config-if)#do write Building configuration... [OK] Router(config)#int gig 3/0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet3/0, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to up Router(config-if)#ip add 172.28.45.18 255.255.255.252 Router(config-if)#do write Building configuration... [OK] Router#show ip int br Interface IP-Address OK? Method Status Protocol GigabitEthernet0/0 192.168.45.22 YES manual up up GigabitEthernet1/0 172.28.45.6 YES manual up up GigabitEthernet2/0 172.28.45.9 YES manual up up GigabitEthernet3/0 172.28.45.18 YES manual up up```
9	Konfigurasi dan tampikan IP router Cabang Semarang(CS)	```Router>ena Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#int gig 0/0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up Router(config-if)#ip address 192.168.45.30 255.255.255.248 Router(config-if)#do write Building configuration... [OK] Router(config)#int gig 2/0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet2/0, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet2/0, changed state to up Router(config-if)#ip address 172.28.45.2 255.255.255.252 Router(config-if)#do write Building configuration... [OK]```

		``` Router(config)#int gig 1/0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet1/0, changed state to up Router(config-if)#ip address 172.28.45.5 255.255.255.252 Router(config-if)#do write Building configuration... [OK] Router(config)#int gig 3/0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet3/0, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to up Router(config-if)#ip add 172.28.45.21 255.255.255.252 Router(config-if)#do write Building configuration... [OK] Router#show ip int br Interface IP-Address OK? Method Status Protocol GigabitEthernet0/0 192.168.45.30 YES manual up up GigabitEthernet1/0 172.28.45.5 YES manual up up GigabitEthernet2/0 172.28.45.2 YES manual up up GigabitEthernet3/0 172.28.45.21 YES manual up up ``` KP ``` Router(config)#router ospf 10 Router(config-router)#network 192.168.45.0 0.0.0.7 area 0 Router(config-router)#network 172.28.45.0 0.0.0.3 area 0 Router(config-router)# 05:54:44: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.45.30 on GigabitEthernet2/0 from LOADING to FULL, Loading Done network 172.28.45.16 0.0.0.3 area 0 Router(config-router)#network 172.28.45.16 0.0.0.3 area 0 Router(config-router)#network 172.28.45.12 0.0.0.3 area 0 Router(config-router)#end ``` CB ``` Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#route ospf 10 Router(config-router)#network 192.168.45.8 0.0.0.7 area 0 Router(config-router)#network 172.28.45.8 0.0.0.3 area 0 Router(config-router)#network 172. 06:11:48: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.45.22 on GigabitEthernet2/0 from LOADING to FULL, Loading Done ^ % Invalid input detected at '^' marker. Router(config-router)#network 172.28.45.20 0.0.0.3 area 0 Router(config-router)#network 172.28.45.12 0.0.0.3 area 0 06:12:31: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.45.30 on GigabitEthernet3/0 from LOADING to FULL, Loading Done Router(config-router)#end ``` CM ``` Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#no route ospf 10 Router(config)#route ospf 10 Router(config-router)#network 192.168.45.16 0.0.0.7 area 0 Router(config-router)#network 172.28.45.4 0.0.0.3 area 0 Router(config-router)#network 172 06:39:33: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.45.30 on GigabitEthernet1/0 from LOADING to FULL, Loading Done ^ % Invalid input detected at '^' marker. Router(config-router)#network 172.28.45.16 0.0.0.3 area 0 Router(config-router)#network 172.28.45.16 0.0.0.3 area 0 06:40:25: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.45.6 on GigabitEthernet3/0 network % Incomplete command. Router(config-router)#network 172.28.45.8 0.0.0.3 area 0 Router(config-router)# 06:41:36: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.45.14 on GigabitEthernet2/0 from LOADING to FULL, Loading Done ``` CS ``` Router(config)#route ospf 10 Router(config-router)#network 192.168.45.24 0.0.0.7 area 0 Router(config-router)#network 172.28.45.4 0.0.0.3 area 0 Router(config-router)#network 172.28.45.20 0.0.0.3 area 0 Router(config-router)#network 172.28.45.0. 0.0.0.3 area 0 ^ % Invalid input detected at '^' marker. Router(config-router)#network 172.28.45.0 0.0.0.3 area 0 Router(config-router)#end ``` Router#show ip ospf neighbor
11	Cek konfigurasi ospf di masing router (CB, CM, dan CS) dengan perintah berikut, tampilkan hasilnya. show ip ospf neighbor show ip ospf database	``` Neighbor ID Pri State Dead Time Address Interface 192.168.45.14 1 FULL/BDR 00:00:31 172.28.45.13 GigabitEth 192.168.45.22 1 FULL/BDR 00:00:36 172.28.45.18 GigabitEth 192.168.45.30 1 FULL/DR 00:00:36 172.28.45.2 GigabitEth Router#show ip osf database ^ % Invalid input detected at '^' marker. Router#show ip ospf database OSPF Router with ID (192.168.45.6) (Process ID 10) Router Link States (Area 0) Link ID ADV Router Age Seq# Checksum Link count 192.168.45.6 192.168.45.6 697 0x80000009 0x007dd0 4 192.168.45.30 192.168.45.30 749 0x80000017 0x00d52b 4 192.168.45.22 192.168.45.22 632 0x80000014 0x001004 4 192.168.45.14 192.168.45.14 629 0x80000013 0x00d340 4 Net Link States (Area 0) Link ID ADV Router Age Seq# Checksum 172.28.45.14 192.168.45.6 535 0x8000000f 0x006d97 172.28.45.17 192.168.45.6 697 0x8000000e 0x00db0a 172.28.45.2 192.168.45.30 1643 0x80000012 0x009eae 172.28.45.5 192.168.45.30 897 0x80000013 0x00dcc2 172.28.45.9 192.168.45.22 632 0x80000004 0x00f7da 172.28.45.21 192.168.45.30 560 0x80000014 0x003c1e Router# show ip route ospf 172.28.0.0/30 is subnetted, 6 subnets O 172.28.45.4 [110/2] via 172.28.45.2, 01:11:01, GigabitEthernet2/0 [110/2] via 172.28.45.18, 01:11:01, GigabitEthernet3/0 ```

	show ip route osfp	``` Router>show ip ospf neighbor Neighbor ID Pri State Dead Time Address Interface 192.168.45.30 1 FULL/DR 00:00:31 172.28.45.21 GigabitEthernet3/0 192.168.45.22 1 FULL/DR 00:00:35 172.28.45.9 GigabitEthernet2/0 192.168.45.6 1 FULL/DR 00:00:39 172.28.45.14 GigabitEthernet1/0 Router>show ip ospf database OSPF Router with ID (192.168.45.14) (Process ID 10) Router Link States (Area 0) Link ID ADV Router Age Seq# Checksum Link count 192.168.45.14 192.168.45.14 759 0x80000013 0x00d340 4 192.168.45.30 192.168.45.30 879 0x80000017 0x00d52b 4 192.168.45.6 192.168.45.6 828 0x80000009 0x007dd0 4 192.168.45.22 192.168.45.22 762 0x80000014 0x001004 4 Net Link States (Area 0) Link ID ADV Router Age Seq# Checksum 172.28.45.2 192.168.45.30 1772 0x80000012 0x009eae 172.28.45.5 192.168.45.30 1027 0x80000013 0x00dcc2 172.28.45.17 192.168.45.6 828 0x8000000e 0x00db0a 172.28.45.9 192.168.45.22 762 0x80000004 0x00f7da 172.28.45.21 192.168.45.30 690 0x80000014 0x003c1e 172.28.45.14 192.168.45.6 666 0x8000000f 0x006d97 Router>show ip route ospf 172.28.0.0/30 is subnetted, 6 subnets O 172.28.45.0 [110/2] via 172.28.45.21, 01:41:52, GigabitEthernet3/0 [110/2] via 172.28.45.14, 01:41:52, GigabitEthernet1/0 O 172.28.45.4 [110/2] via 172.28.45.9, 01:13:09, GigabitEthernet2/0 [110/2] via 172.28.45.21, 01:13:09, GigabitEthernet3/0 O 172.28.45.16 [110/2] via 172.28.45.9, 01:13:09, GigabitEthernet2/0 [110/2] via 172.28.45.14, 01:13:09, GigabitEthernet1/0 192.168.45.0/29 is subnetted, 4 subnets Router#show ip ospf neighbor Neighbor ID Pri State Dead Time Address Interface 192.168.45.30 1 FULL/DR 00:00:32 172.28.45.5 GigabitEthernet1/0 192.168.45.6 1 FULL/DR 00:00:36 172.28.45.17 GigabitEthernet3/0 192.168.45.14 1 FULL/BDR 00:00:32 172.28.45.10 GigabitEthernet2/0 Router#show ip ospf database OSPF Router with ID (192.168.45.22) (Process ID 10) Router Link States (Area 0) Link ID ADV Router Age Seq# Checksum Link count 192.168.45.22 192.168.45.22 895 0x80000014 0x001004 4 192.168.45.30 192.168.45.30 1013 0x80000017 0x00d52b 4 192.168.45.6 192.168.45.6 961 0x80000009 0x007dd0 4 192.168.45.14 192.168.45.14 893 0x80000013 0x00d340 4 Net Link States (Area 0) Link ID ADV Router Age Seq# Checksum 172.28.45.9 192.168.45.22 895 0x80000004 0x00f7da 172.28.45.5 192.168.45.30 1161 0x80000013 0x00dcc2 172.28.45.17 192.168.45.6 961 0x8000000e 0x00db0a 172.28.45.21 192.168.45.30 824 0x80000014 0x003c1e 172.28.45.14 192.168.45.6 800 0x8000000f 0x006d97 172.28.45.2 192.168.45.30 107 0x80000015 0x00f88e Router#show ip route ospf 172.28.0.0/30 is subnetted, 6 subnets O 172.28.45.0 [110/2] via 172.28.45.5, 01:15:32, GigabitEthernet1/0 [110/2] via 172.28.45.17, 01:15:32, GigabitEthernet3/0 O 172.28.45.12 [110/2] via 172.28.45.17, 01:15:32, GigabitEthernet3/0 [110/2] via 172.28.45.10, 01:15:32, GigabitEthernet2/0 O 172.28.45.20 [110/2] via 172.28.45.5, 01:15:32, GigabitEthernet1/0 [110/2] via 172.28.45.10, 01:15:32, GigabitEthernet2/0 192.168.45.0/29 is subnetted. 4 subnets Router#show ip ospf neighbor Neighbor ID Pri State Dead Time Address Interface 192.168.45.6 1 FULL/BDR 00:00:34 172.28.45.1 GigabitEthernet2/0 192.168.45.22 1 FULL/BDR 00:00:38 172.28.45.6 GigabitEthernet1/0 192.168.45.14 1 FULL/BDR 00:00:36 172.28.45.22 GigabitEthernet3/0 Router#show ip ospf database OSPF Router with ID (192.168.45.30) (Process ID 10) Router Link States (Area 0) Link ID ADV Router Age Seq# Checksum Link count 192.168.45.30 192.168.45.30 1150 0x80000017 0x00d52b 4 192.168.45.6 192.168.45.6 1099 0x80000009 0x007dd0 4 192.168.45.22 192.168.45.22 1033 0x80000014 0x001004 4 192.168.45.14 192.168.45.14 1031 0x80000013 0x00d340 4 Net Link States (Area 0) Link ID ADV Router Age Seq# Checksum 172.28.45.5 192.168.45.30 1298 0x80000013 0x00dcc2 172.28.45.2 192.168.45.30 243 0x80000015 0x00f88e 172.28.45.21 192.168.45.30 961 0x80000014 0x003c1e 172.28.45.17 192.168.45.6 1099 0x8000000e 0x00db0a 172.28.45.9 192.168.45.22 1033 0x80000004 0x00f7da 172.28.45.14 192.168.45.6 937 0x8000000f 0x006d97 Router#show ip route ospf 172.28.0.0/30 is subnetted, 6 subnets O 172.28.45.8 [110/2] via 172.28.45.22, 01:17:41, GigabitEthernet3/0 [110/2] via 172.28.45.6, 01:17:41, GigabitEthernet1/0 O 172.28.45.12 [110/2] via 172.28.45.22, 01:46:24, GigabitEthernet3/0 [110/2] via 172.28.45.1, 01:46:24, GigabitEthernet2/0 O 172.28.45.16 [110/2] via 172.28.45.6, 01:17:41, GigabitEthernet1/0 [110/2] via 172.28.45.1, 01:17:41, GigabitEthernet2/0 192.168.45.0/29 is subnetted. 4 subnets ```
12	Cek tracert dan ping ke wali.id dari cabang bandung, Semarang, makasar dan Semarang.	CB-KP ``` C:\>ping wali.ad Pinging 192.168.45.0 with 32 bytes of data: Reply from 172.28.45.14: bytes=32 time<1ms TTL=254 Reply from 172.28.45.14: bytes=32 time=10ms TTL=254 Reply from 172.28.45.14: bytes=32 time=10ms TTL=254 Reply from 172.28.45.14: bytes=32 time=1ms TTL=254 Ping statistics for 192.168.45.0: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 10ms, Average = 5ms C:\>tracert wali.ad Tracing route to 192.168.45.0 over a maximum of 30 hops: 1 77 ms 0 ms 0 ms 192.168.45.14 2 0 ms 0 ms 0 ms 172.28.45.14 Trace complete. ``` CM-KP ``` C:\>ping wali.ad Pinging 192.168.45.0 with 32 bytes of data: Reply from 172.28.45.17: bytes=32 time=38ms TTL=254 Reply from 172.28.45.17: bytes=32 time=10ms TTL=254 Reply from 172.28.45.17: bytes=32 time<1ms TTL=254 Reply from 172.28.45.17: bytes=32 time<1ms TTL=254 Ping statistics for 192.168.45.0: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 38ms, Average = 12ms C:\>tracert wali.ad Tracing route to 192.168.45.0 over a maximum of 30 hops: 1 10 ms 0 ms 0 ms 192.168.45.22 2 0 ms 0 ms 0 ms 172.28.45.17 ``` CS-KP

		 C:\>ping wali.ad Pinging 192.168.45.0 with 32 bytes of data: Reply from 172.28.45.1: bytes=32 time=84ms TTL=254 Reply from 172.28.45.1: bytes=32 time<1ms TTL=254 Reply from 172.28.45.1: bytes=32 time<1ms TTL=254 Reply from 172.28.45.1: bytes=32 time=10ms TTL=254 Ping statistics for 192.168.45.0: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 84ms, Average = 23ms C:\>tracert wali.ad Tracing route to 192.168.45.0 over a maximum of 30 hops: 1 38 ms 0 ms 0 ms 192.168.45.30 2 0 ms 0 ms 10 ms 172.28.45.1 Trace complete.